

Austrian Business Cycle Theory

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Austrian Business Cycle Theory

AN INTRODUCTION

PAUL F. CWWK

MISESINSTITUTE

AUBURN, ALABAMA

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Published 2024 by the Mises Institute

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ISBN: 978-1-61016-773-4

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Preface

THIS MONOGRAPH HAS COME ABOUT because of my inability to find a book like this. When I teach Principles of Macroeconomics, I present the topic from several points of view. Modern textbooks tend to cover Keynesianism, monetarism, and (maybe) New Keynesianism and then to offer a modern synthesis of them. Since the Austrian view of the business cycle is always lacking from these texts, I am compelled to assign a collection of outside readings to supplement my lectures. What I would like to assign, something short and up to date, unfortunately does not exist. This monograph, a product of my lectures to my students, is designed to fill that needed gap.

The purpose of this monograph is to construct a model to illustrate macroeconomics from an Austrian perspective. It begins by laying out the foundational components that will be used in the model. Once the foundation is laid, a basic Austrian model of the macroeconomy is presented. To this model more detail is progressively added to construct an intermediate-level model. Advanced questions critiquing the model follow. The questions show that while the intermediate model represents a good principles-level understanding of how Austrians see business cycles, it leaves open many directions for advanced students to explore. At significant points throughout, the Austrian approach is contrasted with the mainstream. Readings for further inquiry and discussion are provided in the final section.

I want to thank everyone who has helped with this project. In particular, I want to thank Brian Balfour, Robert Batemarco, Michael Carpenter, Mateusz Machaj, Daniella Bassi, and Mark Thornton for

reading and commenting on the manuscript. My deepest thanks go to my tireless editor (but usually tired wife), Heidi, who helps my dreams become real.

Introduction

In the fall of 2009, an economist from the Federal Reserve branch at Charlotte gave a talk at my university explaining the causes and consequences of the 2007–9 recession. He asked us to imagine the economy as a guy on a bicycle. As the bicyclist pedaled down the road, a car knocked him into a ditch. The Fed economist told us that what we needed to do was to get the guy back on the bike and start him pedaling again. It left me dumbfounded. While this was a succinct story, it was not an economic theory. Where did the car come from? Will it come back? Are there other cars? When I asked where these shocks (the cars) come from and how we can avoid them, the answers were “They come from outside of the system” (meaning that nobody, not even economists at the Fed, has any idea where they come from) and “These shocks are unavoidable” (meaning that the government always needs to be ready to leap into action). Despite my questioning, the Fed economist really did not answer these questions. Further complicating the situation, as economists like to do, he dressed up the terminology and talked over the heads of the audience. It is time to strip away the jargon and present a clear theory.

Modern macroeconomics does not actually have a theory of business cycles. In the Fed economist’s story, the car is what economists call “an exogenous shock”—meaning that something from outside of the model negatively impacts the economy. These shocks can be basically anything, from bad weather to new political regimes. One of the main tools used by modern macroeconomists is called the dynamic stochastic general equilibrium (DSGE) model. “Stochastic” simply means

that there is a pattern in the aggregate data, and “dynamic stochastic” means that this pattern tends to occur over time. Anything outside of the model is “exogenous,” and anything that does not fit with general equilibrium is a “shock.” So when economists say that an “exogenous shock” impacted their DSGE model, they are saying that something unknown has disrupted observed historical patterns. In other words, they are basically telling you that they have no idea why this event happened, if it will happen again, or what steps can be taken to prevent it from happening in the future. The car is just gone, so now we need to focus on the bicyclist.

The Austrian School of economics rejects macroeconomics’ atheoretical approach. Austrian economists argue that macroeconomics must be built upon microeconomic foundations. Only once we understand how the smaller parts of an economy work together can we begin to understand how the larger system works.

Part 1

The Foundation of Economic Growth

BEFORE WE CAN UNDERSTAND HOW an economy “goes wrong,” we first need to understand how an economy “goes right.”

The most basic component of any economy is the transaction that takes place between two people. In any free-market transaction, we make two basic assumptions: people are acting voluntarily and property rights are clearly defined. In other words, there is no force, coercion, or fraud. Under these conditions, when two people trade, each side expects to benefit, both psychologically and materially. Under the premise of the double coincidence of wants, each side must place a higher value on what they are trading for than on what they are giving up. For example, if person A has apples and person B has blueberries, then person A must value the blueberries more than the apples *and* person B must value the apples more than the blueberries. If this condition is not met, then no trade will occur. Thus because each trade represents a double coincidence of wants, we must conclude that each trade brings at least a psychological benefit to both traders.

By understanding the benefits that accrue to individual traders, we can by extension see how an entire economic community benefits from trade. Through the law of comparative advantage, we can also conclude that when people specialize in their lowest-opportunity-cost activity and then trade, both sides also become materially better off (see Ayau [2007] for clear examples and an uncomplicated discussion of how this law works). The power of the law of comparative advantage is found in the reduction of costs. When the cost of production is reduced, it is

straightforward to understand how the cost reducer benefits directly. For example, if a person finds a more efficient way to organize his tools, then that person will get his job done more quickly and will have more free time left afterward. Yet how is this benefit passed on to others?

Let us suppose that there is a manufacturing business selling one thousand units a day at a price of \$10 and a cost of \$8 per unit. A quick calculation reveals that the company is making \$2,000 per day (Profit = Revenue – Costs, so for this business, $\$10,000 - \$8,000 = \$2,000$). In order to increase its profits, the company can change its prices, reduce its costs, or try a combination of both. If the company chooses to raise the price of its units, it will lose some customers. If it loses too many customers (due to consumers' high price sensitivity—the demand is elastic), then its total revenue will fall. The company could instead cut the price to attract more customers (due to elastic demand), but the price decrease might outweigh the influx of new customers. In reality, companies are fairly good at balancing these strategies. They know that if they raise their prices, their customers will be attracted to competitors and that if they lower their prices, they will lose revenue. Both choices cut into profits. As a result, when a company tries to increase its profit margin in the real world, it focuses on cutting costs rather than on increasing revenue.

Suppose that our fictitious company finds new ways to make its manufacturing operations more efficient so that it can cut its costs from \$8 to \$6 per unit. The benefits of this cost reduction can manifest in three ways. The most obvious is that higher profits accrue to the owners ($\$10,000 - \$6,000 = \$4,000$). They now have more purchasing power to spend on consumer and investment goods. The second benefit comes in the form of higher wages for workers. In equilibrium, factors of production command prices according to the value that they contribute to production. If workers increase their productivity, then over time their wages will increase and they will have more purchasing power. Finally, the company may recognize that with a lower cost per unit, it can lower its output price (from \$10) and attempt to take some customers away from its competitors. The lower price benefits the consumers directly. In each instance, the firm is able to free up resources by cutting costs. These additional resources lead to an increase in the supply of consumer goods, investment goods, or both.

This process of economic growth through expanding supply has gone by many names, perhaps the most popular being Say's law. Say's

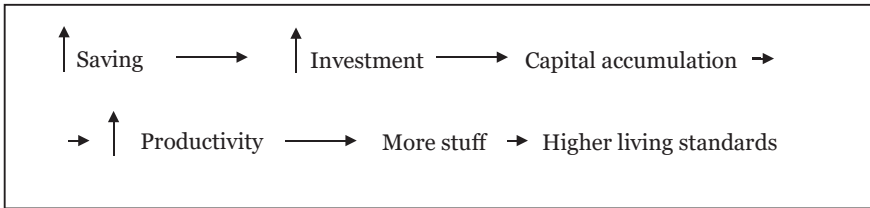


Figure 1: The “magic” formula for economic growth

law starts by pointing out that money is a medium of exchange, a connector of one person’s production and other people’s production. Then it shows that we must produce before we can consume. I have called this idea the “magic” formula for economic growth (of course, there is nothing magical about it; see figure 1).

Let us work through this formula by starting at the end and working in reverse. We can begin by asking, “What is the goal of the economy’s participants?” Generally, we can answer that people would like to become better off (i.e., have a higher standard of living). This conclusion leads to the question, “How can we improve our living standard?” The most generic answer is by gaining more “stuff.” Of course, “stuff” is not limited to physical items. It can include intangibles like free time. However, in order to gain a higher standard of living, one thing is true: we must create *more of something*. But how?

Throughout history, some societies have gained this extra “stuff” through conquest. However, earlier we stated that free-market transactions are free of coercion, so conquest is ruled out as a legitimate way to increase output. Another option is to discover new resources. Untapped resources were discovered and put in use as humanity spread across the globe. However, while some undiscovered resources remain, this option is not really viable, since the occurrence of discoveries is irregular and unreliable. This leaves only one other alternative available—increasing our productivity (i.e., increasing our output using the same amount of time and/or resources or maintaining the same output using less time and/or resources). Without new frontiers and without conquest, new techniques, machines, tools, and equipment are required to increase productivity. In other words, we become more productive through capital accumulation.

Market investments are the source of capital accumulation. Funding is needed to build and establish capital goods and to implement new techniques. Markets are critical to properly allocate investment funds into profitable ventures and to avoid inefficient ones.

The source of investors' funds is savings, also known as *deferred consumption*. Savings comprise more than the money we keep in our bank accounts or in cash. They also include the profits that companies reinvest into their business activities (retained earnings). By choosing not to spend, savers set off a chain reaction. It first impacts consumer prices, pushing them down, and then spreads through the whole of the economy. Simply holding cash has the same effect as other forms of savings.

In summary, the “magic” formula for economic growth shows that an increase in savings leads to an increase in investment. The increase in investment allows more and better tools, machines, and equipment to be used in production (capital accumulation). These tools increase the productivity of workers, which places upward pressure on their wages. The higher level of productivity also means that we can have more goods and services, which are now sold at lower real prices, with the same or fewer inputs. This increase in “stuff” is what allows us to have higher living standards.

The “magic” formula for economic growth focuses our attention on some important conclusions. First, saving is the necessary first step toward economic growth. Even a good idea cannot make an economic impact if there is not the wherewithal, savings, to implement it. Second, contrary to Keynesian theory, we cannot consume our way to prosperity. While consumption is the ultimate goal of all production, we must produce in order to consume; this insight is the essence of Say's law. If an economy consumes capital, instead of accumulating capital through savings, then incomes and living standards will fall. Third, government spending is not necessary to foster economic growth. Economic growth is strictly a private sector activity. Finally, and very importantly, economic growth that follows the “magic” formula leads to gently falling prices. In other words, economic growth that arises from gains in productivity leads to natural deflation and this natural deflation is *good* (conversely, even with a constant money supply, a shrinking economy would experience inflationary pressures; see Hayek [1999, 215]). The idea that an economy needs any level of inflation to grow is simply untrue.

Part 2

Creating a Business Cycle Model

PRELIMINARIES

IN ORDER TO EXAMINE SOMETHING as complex as a macroeconomy, we need to simplify the important variables. A model of an economic system is an abstraction that necessarily excludes all but the most important pieces. Macroeconomists need a simple starting point. The state of equilibrium gives economists a reference point from which they can compare how some economic elements affect, and are affected by, other factors. The assumption of equilibrium means that everything in the model will continue in a balanced state (without change) until something new happens. Embedded in the notion of equilibrium is the idea that entrepreneurs do not make systemic errors. In other words, they tend to get it right. However, a business cycle is characterized by a “cluster of errors” committed by entrepreneurs (see Rothbard 2000, 8). These entrepreneurial errors are spread across the entire economy; they are not simply a gyration within a particular market sector. The source of such widespread error must be something that affects the entire economy.

There are only two elements in an economy that have such a systemic impact: money and time. Money is unique in that, with few exceptions, it touches every trade. When the supply of money increases, the value of every unit falls, though it does not do so uniformly.¹ Because

¹Money is not neutral, meaning that changes in purchasing power are not instantaneous nor even. Changes in value take place at particular points first. The wave of changes is like

of the decrease in purchasing power, more money is needed to make the same purchases. In other words, we see an inflation of prices that affects the entire economy. Money is the prime candidate behind the mechanism that begins and propagates the business cycle.

The other element that permeates the economy is time. Most other economic schools assume that continuous production allows us to discard time as an element in macroeconomic models. A key feature of the Austrian school is its insistence that production takes time. In addition to emphasizing that there is a time structure to production, Austrians consider time as the basis of interest rates. Interest rates are intertemporal prices; that is, prices that connect the present to the future. While time does connect all economic decisions, it alone cannot be the cause of a business cycle because human beings cannot change the flow of time. Nevertheless, interest rates, which reflect our relation to time, play an important part in the swings of the business cycle.

The Basic Idea

The Austrian school states that a business cycle moves through five recognizable stages. These stages (seen in figure 2) are (1) monetary expansion, (2) unsustainable boom, (3) upper turning point, (4) liquidation phase, and (5) recovery.

In the real world, these stages are sometimes entangled and less clearly defined. In fact, due to bad economic policy, an economy might not even experience a true recovery, as was the case of the 2007–9 recession.

A Simple Story

Sometimes, the best way to convey an idea is through the use of an allegory.² Imagine you are in a class and the professor says that all of the grades have been destroyed by a computer virus. This unreasonable professor says that your grade for the entire class now rests on a

honey spreading slowly across a surface as it is poured out. Hayek (2012, 326) used this analogy in a technical debate. For a very readable article on the nonneutrality of money, see Mises (1990b).

²This retelling follows my article “The Economy Pulls an All-Nighter,” which appeared in the *Mises Daily* on September 29, 2003.

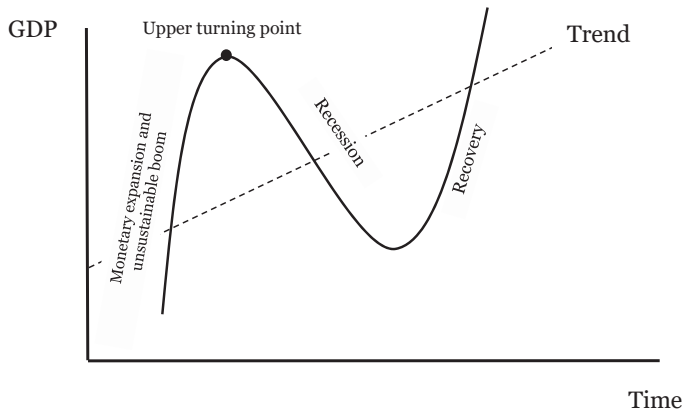


Figure 2: A business cycle

fifteen-page paper that is due on his desk at eight tomorrow morning! Of course, you have not started the paper. In fact, you do not even have more than a basic topic. Your other commitments prevent you from even getting to your computer until after dinner. You are tired and anxious but begin the paper. Around eleven, the adrenaline has worn off and you are feeling tired. What do you do? You brew a pot of coffee and continue. When two in the morning rolls around, you are still not done, yet the effects of the coffee have faded. You are drowsy again. You need another jolt of caffeine. Will the same amount of coffee provide the same amount of stimulation? No, it will not, and so you need to “increase the dose.” By six in the morning, you are barely able to keep your eyes open, but then you remember that you need footnotes and the bibliography! Again you have a choice, so once again, you increase your caffeine intake. Finally at eight, you slam the paper upon the professor’s desk, go home, and crash, falling soundly asleep.

This story is an allegory for the business cycle. When credit is created and pumped into an economy, it is an artificial stimulant for the economy in the same way the caffeine is for the student. The new credit circulates through the entire body of the economy, sending false signals to all entrepreneurs. Entrepreneurs engage in ventures that are built upon the temporary stimulative credit injections. Austrians call these projects “malinvestments.” When the effects of the stimulant wear off, the monetary authorities have a choice to stop or continue. If they

choose to continue, they will need to increase the dose. But eventually, there must be a crash. Either the monetary authority stops stimulating the economy, which results in a credit crunch, or the economy runs out of savings, which results in a resource crunch.

All other economic schools say that anything we can do to delay or avoid recessions is the best path. In contrast, according to Austrian business cycle theory (ABCT), the liquidation process of a recession is a necessary—although painful—step toward recovery. We have been up all night drinking too many sugary, caffeinated frappes. When the student sleeps, the time that the body uses to flush out all of the consumed stimulant and recover from the binge is essentially wasted. No other activities can be done. Similarly, the economy uses the recession to clear out the malinvested capital so that future growth will be on a firm foundation. But even if the economy returns to its precycle levels, the span of time spent in the cycle has also been wasted.

The Intermediate-Level Model

In his book *Time and Money* (2001), Roger W. Garrison presented an intermediate-level Austrian model of the macroeconomy. The goal of Garrison's model is to convey the essential elements of the ABCT. The model comprises three interlocking graphs: the structure of production (SOP), the production possibilities frontier (PPF), and the loanable funds market.³ In the following sections, we will examine each component individually and then integrate them. Then we will apply the model to three scenarios: when people become more patient (meaning that they postpone consumption and make real resources available for investment purposes), when a price ceiling on interest rates is imposed, and when the monetary authorities expand the money supply through credit creation. This final scenario is the Austrian business cycle theory.

The Structure of Production

When something is made, it must go through a distinct process. To make a wooden table, we must first cut down the tree and then take it to a sawmill before it can be transformed into a table and sold to a

³The latter two components are usually found in principles-level textbooks, which makes the concepts more accessible to beginning students.

customer. We cannot cut the tree into boards before we cut the tree down. There is an order to the production process. The SOP represents this order. It shows the degree of capital complementarity between the economy's different production stages.⁴

Of all the goods produced, only a small fraction are for consumption. However, consumption is the sole reason for production. In order to produce, certain elements must be combined. They are labor, natural resources, and time. These three components are called the “original factors of production.” Imagine a survivor alone on an island. Suppose that of all of his needs, the most pressing is hunger. The most basic element needed is the food itself. Without that, there is no chance to alleviate the survivor's hunger. The castaway must work to acquire food. This might entail walking through the brush to get to berry plants or climbing a tree. It also must be recognized that this activity takes time.

If our survivor needs ten units of fruit a day to stay alive and can only gather five units a day, then our castaway will have a short (and probably miserable) life. If ten units can be gathered, then he will be able to live, albeit at a subsistence level. However, if more than ten units can be gathered per day, then the survivor has a choice to make. Suppose that the castaway can gather fifteen units per day. One option is to gather the ten units and then take the rest of the day off (consuming leisure).

Another option is to gather fruit for two days and take the third day off (also consuming leisure). An important alternative arises, however. What if instead of consuming leisure, he does something productive with this time? The castaway might realize that a long pole could be fashioned out of branches to knock the fruit out of the trees, making the job of gathering more productive. The creation of the pole is the creation of a capital good, which can only come about if savings are first accumulated. Capital goods are not directly serviceable, but instead are utilized in the production of finished, directly serviceable goods. There are three kinds of capital goods: capital equipment, intermediate capital (also known as goods in process) and financial capital.

⁴Capital complementarity is the degree to which specific capital goods work together. Opposite to capital complementarity is substitutability, which is the degree to which goods can be used in lieu of another. For further discussion, please see part 3.

Capital equipment is the category that most people think of when discussing capital. The tools, machines, and equipment that enhance our productive capabilities fall into this category. Capital goods are built by combining natural resources, labor, and time (the original factors of production), and often with other capital goods. Capital goods are embedded in specific stages of production. They work cooperatively with other factors of production. In the example of our lone survivor, the pole is capital equipment. It allows him to increase output. Suppose that output is doubled to twenty units of fruit a day. This additional output not only raises our castaway's living standard (more fruit and/or leisure time), but also gives him more time to devote to other projects like building shelter or spear fishing.

When the survivor gathers the fruit, the berries not immediately consumed are called intermediate capital, also known as goods in process. If the berries are used as ingredients in a recipe, these berries become just one part of a greater whole. In a modern economy, most goods and services are goods in process. They are not directly consumable but are necessary components of a final, finished product.

The third type of capital is financial capital. Money is the tool by which all economic actors calculate how best to allocate goods and resources. Monetary calculations take place from the top echelons of financial markets all the way down to the most basic exchange in a farmers' market. Monetary calculation is the tool everyone uses to decide whether to buy, sell, or wait at every level of the economy.

In figure 3, we see that there are four hypothetical stages of production. Production has a specific sequence. If we are building a wooden table, we cannot first manufacture the table and then cut down the tree. Chopping down the tree must come first. As the production process moves forward, value is added. The goods in process become more valuable as they are transformed (e.g., the tree at the sawmill is more valuable than the tree still in the forest). Ultimately, the final output emerges on the right side of the graph as a consumer good (figure 3).

All of the original factors of production (natural resources, labor, and time) are represented in the structure of production. Capital equipment is embedded in each stage of production. As goods in process flow through the SOP from left to right, this intermediate capital is combined with additional resources, labor, and preexisting capital equipment (figure 4). And while the physical goods and services are

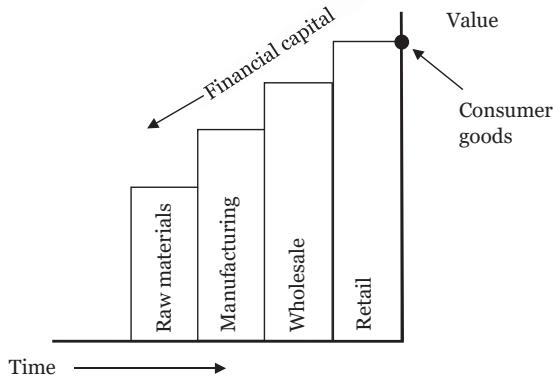


Figure 3: Structure of Production

flowing from left to right, financial capital flows in the opposite direction, from right to left.

We can think of specific forms of capital equipment as being embedded in each stage of production. In our wooden table example, we can imagine people using saws to cut the trees down and trucks to haul them out of the forest. We can imagine the giant cutting equipment at the sawmill.

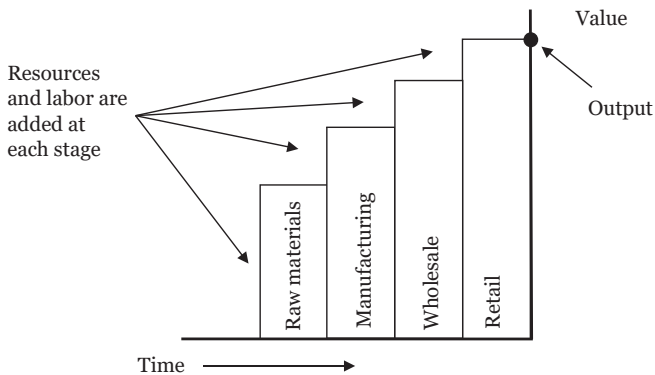


Figure 4: The role of resources and labor in the structure of production

Moreover, there are markets between the various stages of production (see figure 5). These markets are of critical importance, for they illustrate what makes complex production possible. When consumers buy from retailers, they are directly valuing consumer goods. These consumer demands communicate to the various stages of production what to produce and how much. This communication does not take place all at once. It is a step-by-step process. Because of consumer demand, retailers will demand goods from wholesalers. Wholesalers will then demand goods from manufacturers based upon the retailers' demand, which was derived from the consumers' demand. All the demand curves found throughout the SOP are imputed from the consumers' original demand. Only the consumers' demand is a direct valuation, all of the other demand curves in the SOP are ultimately derived from the consumers.

The owners of the apple orchard, the lithium mine, and other original factors of production associate an opportunity cost with the use of their resources. Opportunity costs are the sole basis of supply curves. Thus, supply curves originate with the owners of the original factors of production. These supply curves are then imputed "forward" through the SOP (along the x-axis; see figure 5).

Thus, between each and every stage, the subjective valuations of buyers and sellers interact and form prices. The intermediate capital

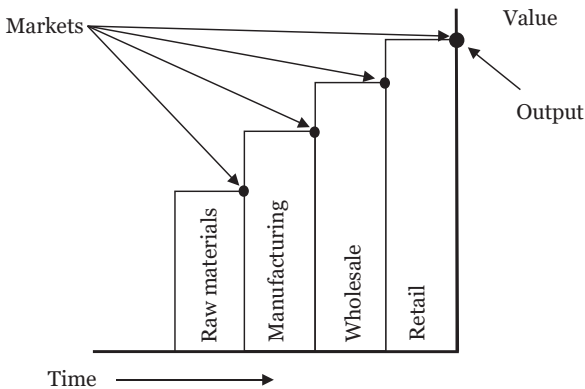


Figure 5: Structure of production with markets

goods flow from left to right, while financial capital flows in the opposite direction.

Of course, there are more than four stages of production in a real economy, which is far more complex than our model. Conceptualizing the SOP as a triangle illustrates the uncountably many stages in a real economy (figure 6).

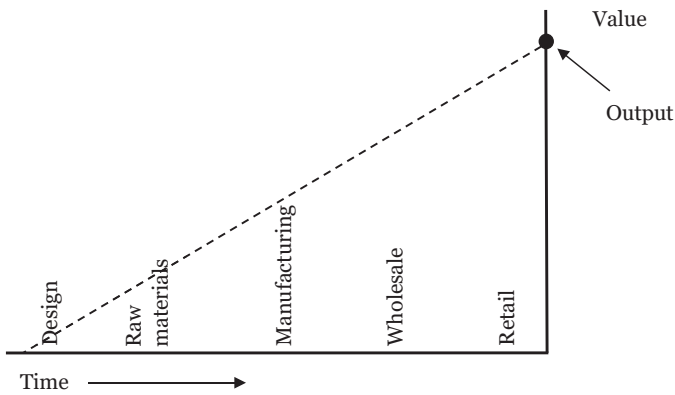


Figure 6: Structure of production as a triangle

Figure 6 lists five general stages, yet it must be emphasized that a particular company or even industry cannot be specifically located on the SOP. A complex economy uses dual-purpose goods and recursive loops. Dual-purpose goods, such as computers, can be used as both consumer goods and as capital equipment. In a recursive loop, a good creates the inputs to make a copy of itself. For example, a steel tool is used to extract iron ore, which is then used to make tools to extract more ore.⁵ While the image of a triangle by no means perfectly captures all of the aspects of a complex macroeconomy, it does capture all of the necessary elements of the SOP.

The Production Possibilities Frontier

The production possibilities frontier (PPF) assumes that, at the present state of technology and with fixed resources, there are two

⁵Recursive loops and other complicating factors will be examined in part 3.

producible goods. Paul Samuelson (1948, 18) created one of the most famous examples of the PPF, showing the trade-off between wartime goods (guns) and civilian goods (butter).⁶ Garrison's (2001) innovation was to replace guns and butter (both consumer goods) by making the trade-off between present goods (consumer goods) and future goods (investment goods). The PPF attempts to illustrate all of the possible combinations for maximum output in the economy.

There are two ways to display the PPF. The first illustrates the maximum *potential* output of an economy, while the second, Garrison's model, shows the maximum *sustainable* output. While Austrians have used both versions, the Garrisonian model, though based on sustainable output, allows the economy to temporarily move beyond sustainable output, which is better for business cycle discussions (figure 7).

If the entire economy were to devote its entire production capacity to consumer goods, then it would produce at point A in figure 7. If the producers shift toward investment goods, not only will there be a reduction in the production of consumer goods, but this reduction will also follow a concave path. The reason the PPF is concave is due to the

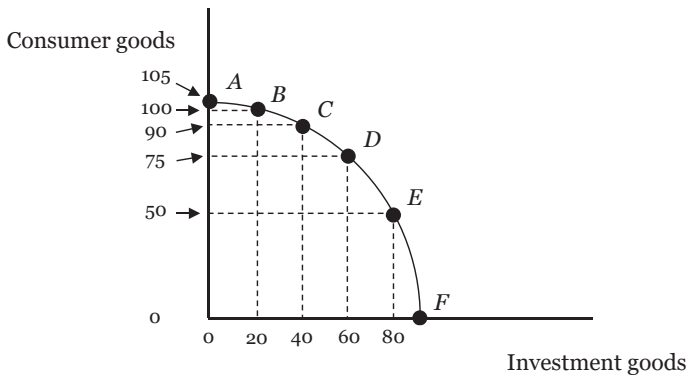


Figure 7: The Garrisonian production possibility frontier

⁶Samuelson (1948, 18) states, "Let us, therefore, simplify (idealize, abstract) and assume that there are *only* two different goods or classes of goods to be produced. For dramatic purposes, let us consider the famous pair, butter and guns, two commodities popularly used to illustrate the wartime problem of choosing between civilian and military production." However, Samuelson did not create the PPF. For example, Gottfried Haberler ([1933] 1959, 176) used a PPF in his discussion about international trade fifteen years earlier.

law of increasing opportunity costs. In order to move from point A to point B , the economy sacrifices five units of consumer goods. When it moves from point B to point C , a larger sacrifice (ten units) is required. As we shift along the PPF, from points A to B to C and so forth, we see that in order to gain equal increases of investment goods, we must sacrifice more consumer goods.

Every point on the PPF represents a possible maximum sustainable output. Any point that is inside the PPF represents an economy below its maximum, generally meaning that it is in a recession. While we can say that being on the PPF is preferred to being in a recession, we cannot make the case that any specific point on the PPF is preferred to another point on the PPF. We can state that more consumption (less saving) today and less investment tomorrow produce less economic expansion. And we can state the inverse, that less consumption (more saving) today and more investment tomorrow produce more economic expansion. However, we cannot say that a particular point is best for an economy, despite our personal preferences.

A point beyond the PPF is possible, but it is not sustainable, in the same way that a person can stay up all night working on an economics paper (occasionally “pull an all-nighter”) but cannot stay up *every* night. Eventually, that person will need to sleep.

Economic growth is illustrated by an expansionary shift of the PPF. In figure 8, we begin at point A_1 , which represents a fully employed economy. At point A_1 , the economy produces C_1 consumer goods and I_1 investment goods. Any increase of consumer goods must come at the

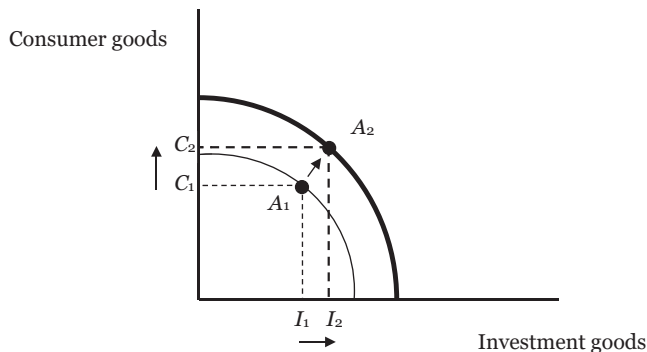


Figure 8: PPF shift showing economic growth

expense of investment goods, unless we can change the PPF boundary. The PPF expands with economic growth, such as when savings and investment fuel an increase in technology that makes more production possible. The expansion is shown by shifting the entire curve upward and to the right. While any point on this new frontier is superior to the old frontier, point A_2 shows that with the use of this new technology we can potentially have both more consumer goods and more investment goods.

The Loanable Funds Market

The loanable funds market shows an economy's aggregate rate of time preference. Time preference reflects the fact that people prefer sooner to later, *ceteris paribus*. If offered the choice between \$100 now and the promise of a \$100 a year from now, people will prefer the \$100 now because it can be used now.⁷

To model time preference, we are going to make a significant, simplifying assumption about the loanable funds market: there is only one interest rate. In our market, the interest rate is determined by time preference. Although in non-Austrian constructions of the loanable funds market the demand is also determined by subjective time preferences, the supply side is often determined by objective productivity factors. In contrast, the Austrian model of this market is premised on the theory that both the demanders and the suppliers of loanable funds are motivated only by their time preferences. This model is called the pure time preference theory (PTPT) of interest rates.

The demanders of loanable funds are those who wish to borrow. Some people have a high time preference, meaning that they are not very patient. These borrowers need the funds quickly and cannot delay their borrowing. Other borrowers have enough options, so they can wait for better rates (low time preference). A downward-sloping

⁷Some people might argue against the idea that time preference always has a positive rate by stating that they prefer lunch at lunchtime and not now. This example violates the *ceteris paribus* restriction. The *ceteris paribus* caveat means that we are assuming that all other factors are held constant. When a person says that they would rather wait until lunchtime to have lunch, they are not holding all other factors constant. A rather important factor, one's hunger, changes between now and lunchtime.

demand curve captures the entire array of time preferences of all those who wish to borrow loanable funds.

The suppliers of loanable funds are savers—those who are deferring consumption. The upward sloping supply curve captures their entire array of time preferences. At the lower portion of the supply curve are those who are natural savers. They need little inducement to keep them from spending. At the higher portion of the supply curve, we see people who love to spend. In order to induce them to defer their consumption, a large reward must be offered. Thus, as interest rates rise, we see an increase in the quantity of loanable funds supplied. The supply curve slopes upward.

Putting the supply and demand curves together, we create a model of the loanable funds market (figure 9). The vertical axis measures the intertemporal price that emerges—the interest rate. Along the horizontal axis, we see the quantity of loanable funds supplied and demanded

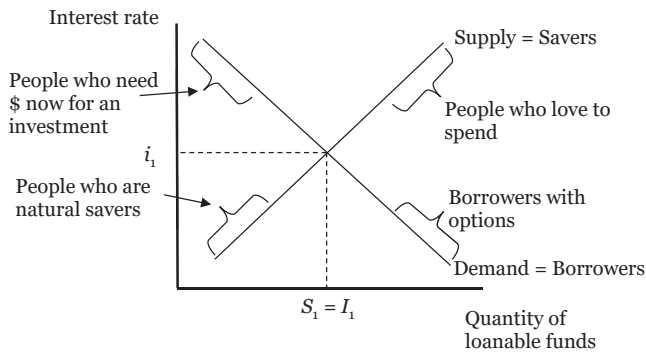


Figure 9: The loanable funds market

in the market. In equilibrium, there is an interest rate (i_1) that balances the quantity supplied (savings, S_1) with the quantity demanded (investment, I_1).

The result is a market that reflects the time preference of its society. As indicated above, other schools of economics include the objective factor of productivity, but the Austrians do not.

The question “Why not include productivity?” needs to be examined. First, a distinction between rents and interest returns must be

made. Every factor of production earns a return—a rent. This rent is the price that must be paid to the owner of a factor of production. In equilibrium, every factor earns a return (rent) that is equal to its marginal product. The marginal productivity of a piece of capital equipment explains the height of its rental price. However, the marginal productivity does not explain why these rents should be discounted across time. A change in the productivity of a capital good will change its rent, but not the rent's discount rate across time (the rate of interest). The discount rate is separate and independent from the productivity of the capital good. Thus, interest rates are based upon time preference, and rents are based upon a factor's productivity.

Putting the Models Together

The Garrisonian model of a macroeconomy interlocks the three preceding graphs (figure 10).⁸

We begin in the lower right, with the loanable funds market. In this market, savers and borrowers come together. At interest rate i_0 , the quantity of loanable funds supplied (S_0) matches the quantity of loanable funds borrowed and invested (I_0). It is important to emphasize that the amount of investment depends upon the amount of savings; saving is the first step in the “magic” formula for economic growth.

The amount of funds used to create investment goods (I_0) is connected to the PPF by the corresponding horizontal axes. In equilibrium, the economy is at full employment, so we find the economy at point A_0 on the frontier. At point A_0 , the corresponding amount of consumption goods produced is C_0 . Finally, C_0 links with the SOP (in the upper left). The slope of the SOP reflects the rate of time preference (the interest rate) in the economy. It is from this equilibrium that we start our macroeconomic analysis.

⁸Unfortunately, it is a static model, meaning that it captures a snapshot in time. As we use the model, we need to keep in mind the dynamic nature of these elements.

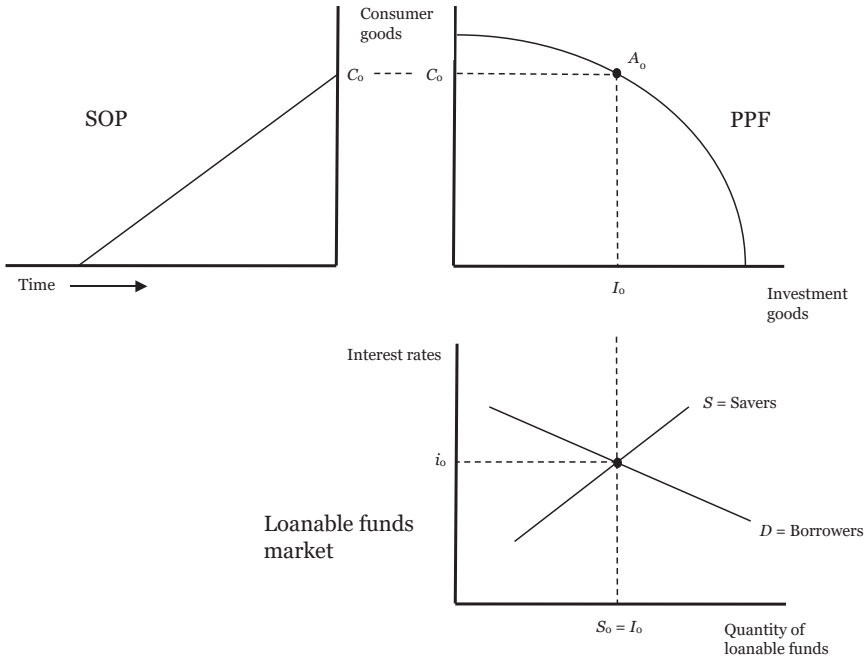


Figure 10: The Garrisonian macroeconomic model

SCENARIO 1 PEOPLE BECOME MORE PATIENT

Now let us put this model into action and demonstrate how it works. Suppose that people become more patient. In other words, they become more willing to save and to defer consumption. The first implication of this change will be found in the loanable funds market. The supply of loanable funds will shift to the right, lowering the equilibrium interest rate and increasing the quantity of loanable funds saved and borrowed (figure 11).

With more borrowed funds allocated toward the construction and implementation of investment goods, we move along the PPF. There is an increase in investment goods and a decrease in consumer goods.

Note that this conclusion completely differs from mainstream economic analysis. Mainstream economic analysis looks at gross domestic product (GDP), which is calculated by adding the following components: consumption spending, investment spending, government spending, and

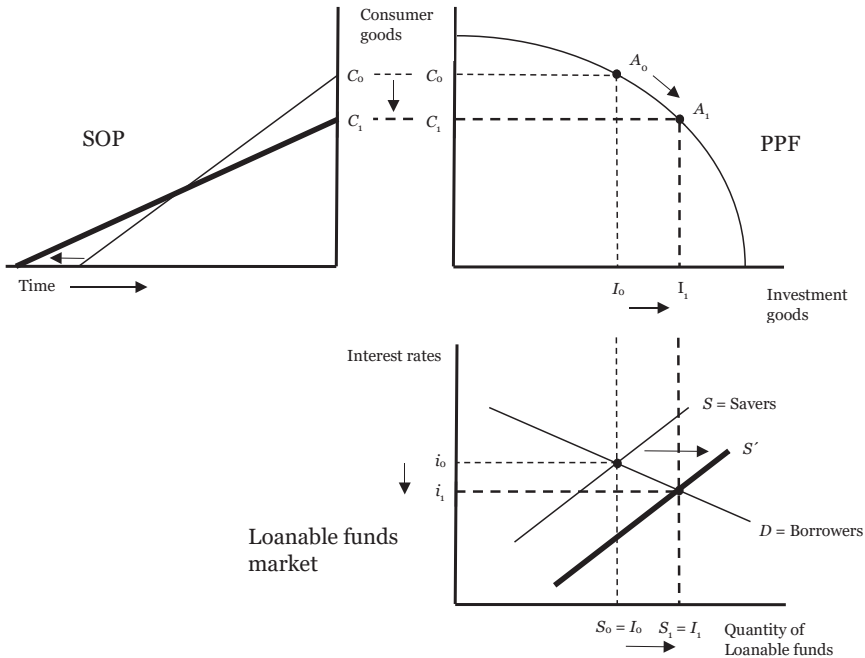


Figure 11: People become more patient

net export spending. Consumption spending makes up about two-thirds of the total GDP. In the mainstream models, the increase in investment spending does not make up for the decrease in consumption spending, so the GDP statistic falls with less consumer spending. Since the GDP statistic is the prime measure of the well-being of the economy not only for economists but also for politicians and journalists, there is a strong political incentive to prevent GDP from falling.

As we move along the PPF, we see that even though the whole economy remains at full employment, there are changes in many sectors. These changes are visible in the SOP (figure 12).

With a reduction in demand for consumer goods (because people are becoming more patient) comes a decrease in demand for late-stage goods (all those goods that are closer to the consumers). While demand for consumer and late-stage goods is falling, savings are being borrowed and employed in more complex production patterns. While an expansion of investment goods can take place at any and all points

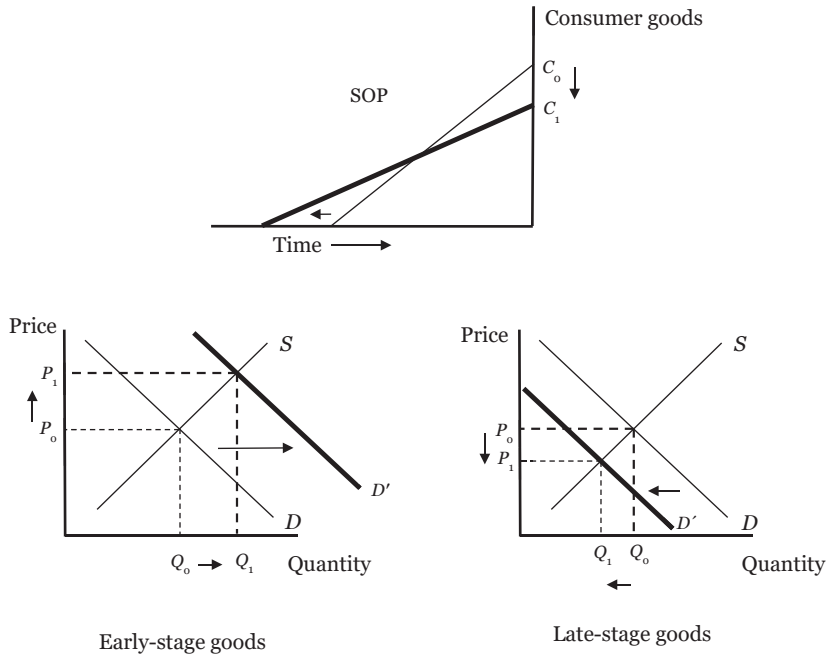


Figure 12: Changes in the structure of production

in the current SOP, the addition of production complexity is illustrated by a lengthening of the structure of production.⁹ We can see the micro-economic effects of these changes in figure 12.

What does lengthening the SOP mean, and why does the SOP lengthen when there is an increase in savings? To describe how economic growth comes from the addition of capital goods, Austrians have used several terms, including higher degrees of roundaboutness, increasing complexity, capital deepening, and lengthening the SOP. The idea begins with the observation that adding complexity to a production process for its own sake is wasteful. The only reason to make a production process more complex is because it yields a benefit; namely, higher profits. Increasing complexity can manifest as a faster

⁹Rothbard (2009, 517–27) demonstrates how the increase of net investment necessarily lengthens the SOP.

assembly line, higher-quality products, more diverse products, and so forth. When a car was designed in the past, draftsmen were employed to draw the plans for each part of it by hand. Any changes to the plans would require many labor hours of redrawing. Today, these changes are made very quickly using computer programs. So while we have the ability to make changes to designs much faster, the process is actually more complex. Someone had to engineer the design program. Someone had to build the computers. Someone had to design the computers. Many stages of production necessarily precede the modern draftsman's work. These extra stages make the production process more complex—longer—but they also make it more efficient.

Decreases in consumption free up resources to shift from one sector of the economy to another. This freeing up of resources grants entrepreneurs the time and wherewithal to engage in new productive projects. These projects—new stages of production—add complexity to the economy. To illustrate these additional stages of production, we graphically extend the economy's entire SOP as seen in figure 12. However, this extension does not mean that any specific economic sectors (like forestry) will always expand as people become more patient.

The model that we are using is called a comparative static model, meaning that it only captures points in time. However, the process that we are attempting to describe takes place through time. One of the key results that was left out of figure 12 is that savings and investment lead to economic growth. To illustrate economic growth, we need to add another step to the comparative static model. As investors use saved funds to lengthen the SOP, the gains in productivity will push the PPF outward (as seen in figure 13). As described above, these economic improvements benefit the entire society. The SOP will shift outward along with economic growth but not smoothly. It will move in a piecemeal fashion. Some sectors will expand while others lag. Furthermore, as new business ventures begin, there will also be an increase in the demand for loanable funds. Initially, the interest rate will fall as people become more patient. However, as more investment projects become feasible, there will be an increase in the demand for loanable funds, shifting the demand curve from D to D' . In figure 13, we are supposing that the interest rate returns to i_0 , but this is just to keep the graph clean. There is no a priori method to determine where the final-equilibrium

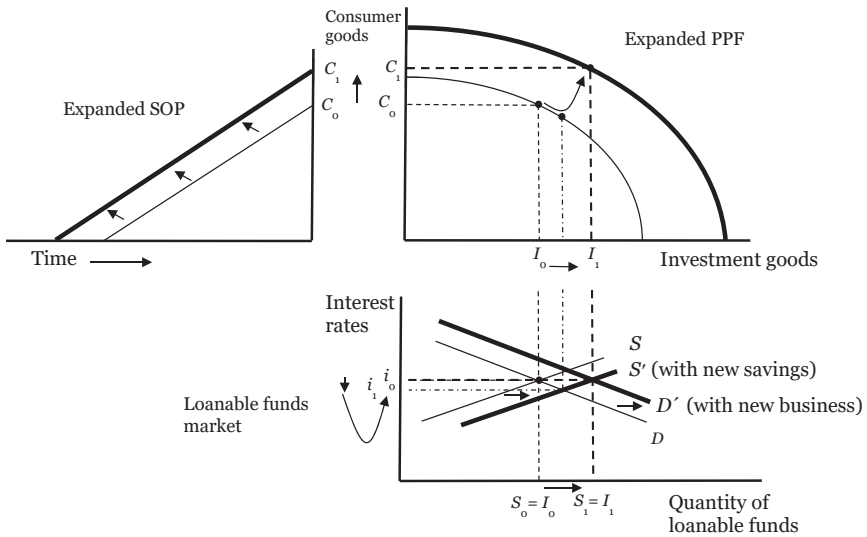


Figure 13: The macroeconomy with economic growth

interest rate will be. However, we can say that higher savings rates will push the PPF out at a faster rate than lower savings rates.

One final point: We are using a comparative statics model, which captures only moments in time, to show economic growth; however, there is no resting equilibrium for economic growth. What we are attempting to describe using static models is a dynamic process.

SCENARIO 2 IMPOSING A PRICE CEILING ON INTEREST RATES

Suppose that the government has decided to place a ceiling on interest rates. As seen in figure 14, if the ceiling is placed below the market interest rate, a shortage of loanable funds will emerge. The lower interest rate sets in motion two reactions. Savers reduce their savings (from S_0 to S_2 ; we move along the supply curve) and increase their consumption (from C_0 to C_2). Entrepreneurs, on the other hand, see a lower cost to borrowing and thereby attempt to increase their borrowing (from I_0 to I_D ; we move along the demand curve). This attempt to borrow more creates a gap between the quantity of loanable funds that borrowers

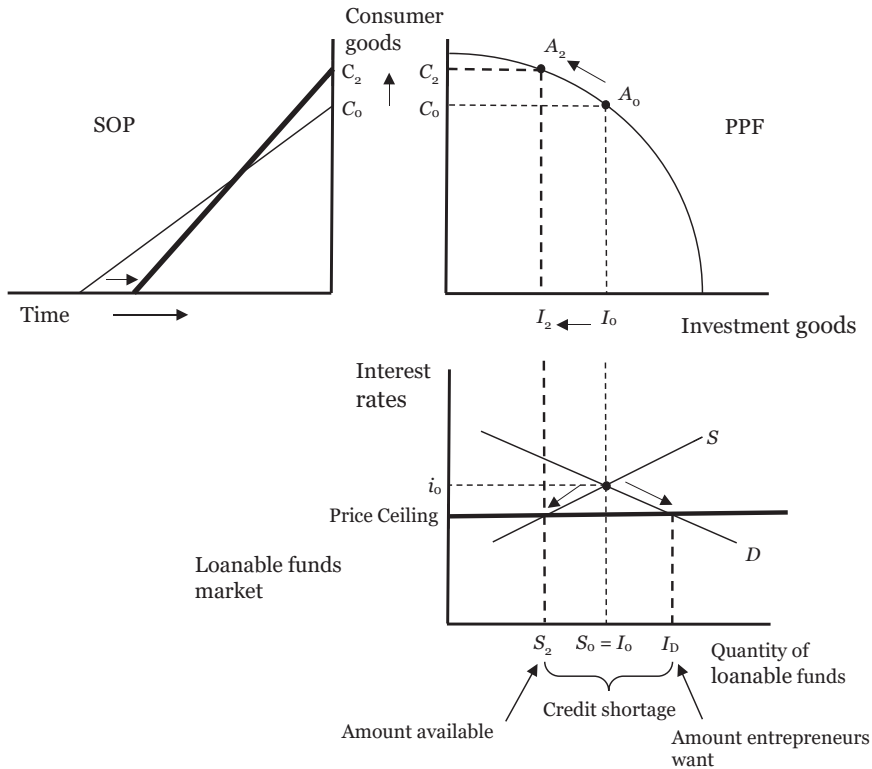


Figure 14: A price ceiling on interest rates

would like and the amount actually saved and available. The result is a credit shortage.

The credit shortage does not allow the investors to borrow as much as they would like. In fact, with a lower savings rate, the amount actually borrowed and invested *decreases* (from I_0 to I_2). We see the impact of the change in the PPF. The equilibrium point moves counterclockwise along the frontier (from A_0 to A_2). Although the economy is still at full employment, the underlying conditions have changed. People are consuming at a higher rate (C_0 has shifted to C_2) and the SOP has become less complex. In effect, what we get is a bust without a boom.

SCENARIO 3 THE BUSINESS CYCLE

Stage 1: Monetary Expansion

Austrian business cycle theory (ABCT) begins with assuming the economy to be in a harmonious equilibrium state. In other words, there is a balance that will continue until something changes. This change can be endogenous (like a simple change in tastes and preferences) or exogenous (like a new government policy). As pointed out above, Austrian economists postulate that money is the only ubiquitous economic element that can cause a cycle.¹⁰

The business cycle begins with an expansion of the money supply. There have been times in history when the money supply has expanded either through conquest or through the discovery of new deposits of precious metals. The business cycles caused by these expansions depended upon the size and duration of the influx of new money. Today, no country uses precious metals as its money, and these older roots of monetary expansion are no longer a concern. Today's money is called "fiat money." It is money that is backed by nothing other than the promise of a government.

The ABCT posits that a monetary authority, usually a central bank, engages in a policy of monetary expansion. While the reasons for this policy range from misguided economic theory to political gains, the reason *why* the central bank adopts an expansionist monetary policy is not relevant to our model. It is the fact that the central bank expands the money supply that matters.

Under modern monetary systems, the typical method to expand the money supply is to create and inject credit into the banking system. While the particulars may differ with location and time, the effects tend to be the same: credit expansion has a great impact on the interest rate markets. The creation and injection of credit causes the supply of loanable funds to shift to the right (S to S' , as seen in figure 15). The effect of adding new money to the market is a reduction of the interest rate. The

¹⁰Some have argued that widely used goods such as oil or energy can cause a business cycle. Despite how widely used such goods are, they cannot be the cause of a systemic cycle. A reduction in the supply of widely used resources will certainly cause disruptions and shift resources, but a cycle exhibits a systemwide up and down.

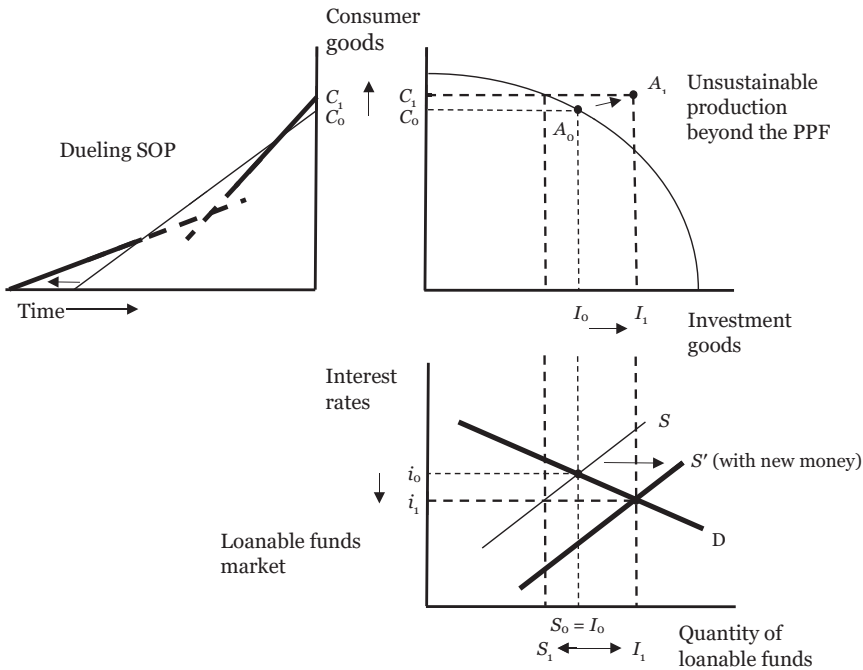


Figure 15: *The unsustainable boom*

lowering of the interest rate is a movement along the demand curve: at this new and lower rate, more investors want to borrow money and finance new projects (I_0 shifts to I_1). Yet what cannot be overlooked is that the addition of new money does not lower the time preferences of savers. On the contrary, as those who are saving (deferring consumption) watch interest rates fall, they will save less (S_0 shifts to S_1). As savings fall, consumption will increase. Normally, when both the interest rate and the quantity of loanable funds available fall, we see a credit shortage develop (as we did in scenario 2). However, in our scenario, this credit shortage is covered by the creation of new credit. This credit comes from nothing. It is literally typed into existence on computers. As a result, we see a separation emerge between actual savings and the amount invested.

Stage 2: Unsustainable Boom

The interest rate is an intertemporal price that connects today with tomorrow. In January 2001, the effective federal funds rate—the major interest rate that the U.S. central bank targets—was at 5.98 percent. Suppose that an entrepreneur, Sam, was considering a business plan projected to yield a 4.00 percent return. At that interest rate, Sam would have shelved the project and simply put the money in a savings account to gain the higher rate of return. Now suppose that the central bank was to engage in an expansionist monetary policy, which reduces the interest rate. In December 2003, for instance, the effective federal funds rate fell to 0.98 percent. Now what would Sam decide if given the same choice? In this case, it is not only rational for Sam to begin the project, but it is his fiduciary duty to pursue the highest rate of return for his business.

The effects of this rate change are far reaching. The interest rate is sending a false signal to entrepreneurs across the whole economy. It is telling entrepreneurs throughout the entire SOP to begin new capital projects which otherwise would not have appeared worth pursuing.

These projects that entrepreneurs are misled into starting are called *malinvestments*. There is no a priori method to determine which particular projects are malinvestments and which are not. This is because money is fungible, so there is no method to determine which money is the newly created credit and which existed before.

At the same time as a malinvestment boom begins (as investment in figure 15 shifts from I_0 to I_1), there is a boom in consumption (an increase from C_0 to C_1). As a result of the increase in both consumption and investment, a shift to a point outside of the PPF (from point A_0 to A_1) occurs, representing an unsustainable boom. This boom is similar to how a person is able to stay up all night but not every night.

We can conclude that the point outside of the PPF is unsustainable by examining the effects of the monetary expansion on the SOP. A decrease in interest rates resulting from credit expansion causes the general public to change its savings-consumption ratio. A lowering of the interest rate encourages people to consume more and save less. The SOP reflects people's increasing consumption by shifting upward, which steepens its right segment, the segment closest to the consumers (the shift creates the partially dashed line starting at point C_1 in

figure 15). This steepening of the right segment of the SOP shows the change in the savings-consumption ratio. The dashed portion of the shifted right segment indicates the direction people's new preferences would lead if they were able to be fully implemented. In other words, consumers are less patient and want more consumer goods now. Only a shorter, less complex, SOP can accommodate these desires.

At the same time that people are consuming more, investors are building more complex capital structures in the economy. They are receiving (false) signals that indicate the general public has more patience to wait for more consumer goods to be produced in the long term (as seen in scenario 1). The entrepreneurs incorrectly believe that they can build a more complex economy and lengthen the SOP. The SOP segment on the left shows the entrepreneurial expansion of building projects. This segment has flattened to indicate the entrepreneurs' (false) perception of the capital structure. The dashed portion of the shifted left segment shows the direction of the SOP if it were to be fully built out under these conditions.

When the two segments of the SOP fail to align, a "dueling structure of production" results.¹¹ The SOP is in disequilibrium and, consequently, is not in a state of rest, meaning that something must change. The boom is unsustainable.

Stage 3: Upper Turning Point

A famous business cycle example—used by Mises—begins with a master builder whose task is to erect buildings using a finite amount of building materials.¹² Suppose that this builder is using bricks to build houses. If the builder has enough bricks to complete four houses, can he begin to build six houses? Yes, but there are not enough bricks to complete all the houses (see figure 16).

In Mises's example, the lower interest rate tempts the builder to start more buildings than he is currently capable of finishing. The interest rate misguides the builder into believing that consumers are more patient—reducing their demand for consumer goods, which frees up resources for investment goods—than they are. To complete

¹¹John P. Cochran (2001, 19) was the first to use the phrase "dueling production structures."

¹²See Mises (1998, 557).

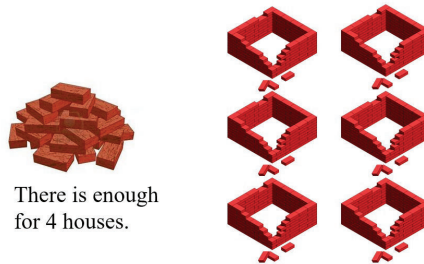


Figure 16: Malinvesting capital inputs

all of the homes after he has started them, the builder will have to get more bricks. Unfortunately, this particular builder is not alone. The situation is the same for all producers. While the shortage of credit was covered through a false expansion of the money supply, the shortage of real resources is genuine. The supply of savings (represented by the bricks that it has been used to purchase) has not increased (as was miscommunicated to the builder) but has actually fallen. Upward pressure is placed upon the price of bricks, one of the input goods for the buildings. As prices rise, entrepreneurs with partially completed projects look for additional financing to meet these higher costs. Interest rates rise.¹³

The economy experiences upward movements of both the interest rate and input prices.¹⁴

We are at the upper turning point. The entrepreneurial “cluster of errors” has been exposed. Entrepreneurs were misled by an artificially low interest rate. The interest rate was artificially low because of expansionary monetary policy, a decision made by nonmarket participants

¹³Please see the section “Unresolved Problems in the ABCT” below for an explanation of how this process necessarily leads to an inverted yield curve.

¹⁴The change in the cost of inputs is more impactful than the increase in interest rates. The increase in the cost of inputs has a one-to-one impact, meaning a 25 percent increase in input prices leads to a 25 percent increase in the amount of financing required. For example, if my cost of bricks increases from \$100 to \$125, I need an extra \$25 to finish my production project. However, a 25 percent increase in interest rates (from 4 percent to 5 percent) has an impact of much less than 25 percent. As the crunch intensifies, 15–20 percent of the collapse in profit margins around the economy is due to the increase in interest rates, but *approximately 80 percent is due to the rise in input prices*. See Cwik (2008).

(central bankers and politicians). The entrepreneurs miscalculated because of the actions of central bankers and politicians.

At the upper turning point, the monetary authorities are in a dilemma and have a choice to make. They can address either the issue of the increasing interest rate or the issue of increasing prices, not both. To stem the increase in interest rates, the monetary authorities can accelerate the rate of monetary expansion—as the student increases the amount of caffeine to finish the paper—or the central bank can address the increasing prices by slowing or halting monetary expansion. A reduction in monetary expansion will result in even higher rates of interest, exacerbating the builder's situation. Politically, accelerating the monetary expansion is the preferred alternative. The reckoning can be delayed, and the consequences of the expansionary policy can be postponed. Regardless of the choice taken by the monetary authorities, the economy is in a disequilibrium, a boom which cannot last forever. Whichever choice the central bank makes, it will not achieve a “soft landing.” Recession is inevitable.

Stage 4: Liquidation (Recession)

If the monetary authorities continue to accelerate the rate of monetary expansion, the economy will eventually run out of bricks (savings). The result will be a real resource crunch. If the monetary authorities decide to stop increasing the interest rate, the result will be a credit crunch. Each of these paths leads to an economic crisis. Either the builder sees input prices rising and can no longer afford to complete the project, which results in the project's termination and the components' liquidation or the builder sees the interest rate become so high that the project is no longer viable, which results in its termination and the materials' liquidation. In August 2006, the federal funds rate climbed to 5.25 percent. The result was the 2007–9 recession. If Sam, the entrepreneur we met in stage 2, had seen this high rate, he would have realized that his project, which only yields a 4 percent rate of return, was no longer viable and must be discontinued.

During the liquidation stage, those firms that invested during the malinvestment boom begin to suffer economic losses. In figure 17, we see the path of a recession. As businesses pull out of the loanable funds market, either by choice or due to an inability to continue operating (from losses), we see a leftward shift in the demand for loanable funds

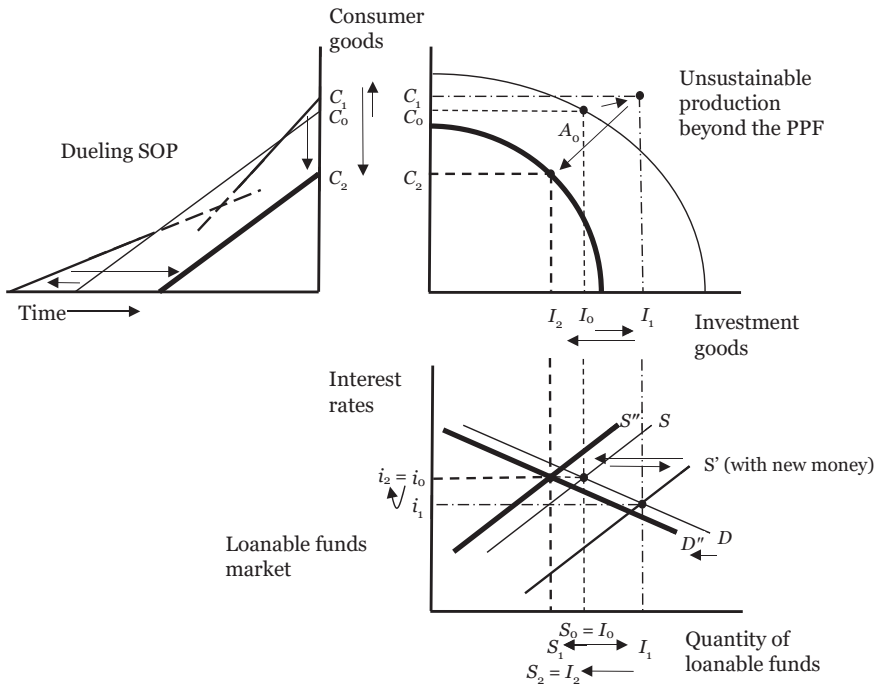


Figure 17: The recession

(D to D''). At the same time, we see that savers are also pulling out of the market (S to S''). Many former savers are now unemployed, as projects have been canceled, while other former savers are businesses who no longer have the funds to reinvest into the company. Note that the interest rate (i_2) is shown to be equal to the initial rate (i_0), but this is merely to keep the graph from becoming too cluttered. There is no a priori reason why the interest rate should return to the same level it was before the monetary expansion. The important characteristic of the loanable funds market in this stage is the level of savings. Savings are the floor upon which the level of investments falls (I_1 to I_2).

We can track the decrease in investment goods in the PPF (also shifting from I_1 to I_2). As people lose their jobs, we see a fall in consumption goods (from C_1 to C_2). As many capital projects are halted, many capital goods that cannot be repurposed are thrown away. Their productive capacity is lost. The entire sustainable productive capacity

of the economy shrinks and the PPF contracts to a lower level. Finally, we see the contraction of the SOP. In figure 17, the new SOP is parallel to the original, but note that this is only because of the assumption that the new interest rate (i_2) matches the original (i_0). There is no a priori reason for the two SOPs to be parallel.

The liquidation phase is a painful, yet necessary, consequence of the preceding expansionist monetary policy. The economy has been pushed into an unsustainable boom and requires correcting. A recession is the process by which an economy converts malinvestments into productive capital, which becomes the foundation for future economic growth. Only the Austrian school argues that liquidation is a necessary condition for recovery.

In order for the economy to recover, the liquidation of failed companies must be completed. As companies fail, their assets are sold and the proceeds are distributed to the creditors and the owners. When these assets are sold, they are sold to other firms at a discounted rate. Because these firms have purchased the capital equipment at a discount, they now have the ability to turn an economic profit, even at the previous prices. For example, suppose that in the 2007–9 recession, the U.S. government truly allowed General Motors to go bankrupt. If GM had been forced to liquidate its assets, it would have had to sell its brands and divisions. There was still value in Corvette and Cadillac. Who would have bought these brands? Perhaps Toyota, BMW, or Elon Musk. It does not matter. A bidding process would have taken place, and one of the bidders would have bought Cadillac. However, the buyer does not have to pay full price in a liquidation sale. The buyer could have gotten Cadillac for pennies on the dollar. The implication is that Cadillac's new owner could turn on the same machines and produce the same cars as GM did. However, since the new owner has a lower cost structure (having bought the division at a discount), the new owner can make a profit where the old owners could not.

This liquidation process is how malinvestments are converted into new fixed capital equipment. Unfortunately, not all malinvestments can be converted back into productive assets. Those assets that cannot be converted are thrown away. Their productive capacity has been wasted and can never be recovered. Additionally, even if the economy ends up at the same starting point, time has been lost, which is also unrecoverable. Finally, many people have lost jobs and suffered through finan-

cial, housing, and food insecurity. These hardships are real and have a human cost that is simply not calculable. They are the main reason why business cycles should be avoided.

Stage 5: Recovery

The liquidation process is necessary for normal economic growth to resume. There is no categorical difference between a recovery and normal economic growth. Both must follow the “magic” formula for economic growth.

Some have argued that Austrian economists have no policy prescriptions to offer. This notion is untrue. The most important policy recommendation is to not start down the path of monetary expansion. Expansionist monetary policy starts the artificial boom and the consequent collapse. If the monetary expansion can be avoided, then business cycles can be prevented.

What can be done once we have begun a business cycle? An important point to recognize is that the boom and the bust do not have to be symmetrical, meaning that the size of the bust does not have to equal the size of the boom. Governments can make the bust much worse than it has to be (as seen in the 1929–39 depression). Governments can make the recession longer than it has to be (as seen in the anemic recovery in the U.S. after 2009 and in Japan during the 1990s). Alternatively, governments can accelerate the recovery (as seen in the 1920–21 recession, discussed below).

Governments often make bad policy choices when dealing with a recession. Typically, to prevent an economic downturn, the monetary authority will accelerate the rate of monetary expansion.¹⁵ This step does not cure a recession; instead, it delays the consequences of the artificial boom, swells the number of malinvestments in the economy, and usually makes the eventual recession worse. Furthermore, governments also tend to prop up failing businesses, thereby preventing the liquidation process. The spending used to prop up failing businesses shifts resources away from more productive firms and into less productive ones. By

¹⁵While it is important to note that the monetary inflation should stop, deflating the money supply to its previous level is also not a solution. Mises (1990, 76) said such a policy would be like after running a person over and then, to fix it, backing the car over him again.

preventing the liquidation process, the government locks the economy into an unproductive capital structure. Governments should recognize that the malinvestments need to be liquidated as quickly as possible. Instead of locking the economy into an unproductive capital structure, governments should make it easier for businesses to conduct mergers and acquisitions. By making the capital structure more malleable, governments allow resources to be shifted into more productive arrangements—creating a strong foundation for economic growth.

Governments also pay people to not work, in the form of unemployment compensation. While these policies might seem humane, they extend unemployment. People delay taking an available job, hoping to find the “perfect” job. Monetary expansion, propping up failing businesses, and unemployment compensation were all enacted in the Great Depression, in the recession of 2007–9, and throughout the 1990s in Japan. In contrast, when a recession came about in January 1920, Congress actually shrunk the size of the federal budget.¹⁶ The result was that resources that were previously consumed by the federal government were released for use in the private sector.

The key to economic recovery (and growth) is the amount of savings. The more savings, the faster the recovery. Policies that discourage savings prolong and deepen the recession. There are two very basic policies that can increase the amount of resources available for the private sector: cutting the size and scope of the federal budget and cutting the amount of tax revenue collected. As we have seen, when the government took a step back in the 1920–21 recession, resources were freed up to be used by the private sector. It was like adding savings directly into the economy.

¹⁶The Democrats had lost control of the House and Senate in the 1918 midterm elections. Furthermore, President Woodrow Wilson had suffered a stroke in September 1919. When the 1920 recession hit in January, Wilson was unable to oppose the budget-balancing cuts in federal expenditures imposed by the new Republican majorities. Federal revenue (*R*) and expenditures (*E*) for the years 1918–22 were as follows: \$3.6 million (*R*), \$12.7 million (*E*) in 1918; \$5.1 million (*R*), \$18.5 million (*E*) in 1919; \$6.6 million (*R*), \$6.4 million (*E*) in 1920; \$5.6 million (*R*), \$5.1 million (*E*) in 1921; \$4.0 million (*R*), \$3.2 million (*E*) in 1922 (“US Federal Budgetary Spending by Year,” Polidiotic, last updated March 14, 2023, <http://www.polidiotic.com/by-the-numbers/us-federal-deficit-by-year/>). Congress also engaged in a series of tax rate cuts, reducing the top marginal rate from 73 percent in 1920 to 58 percent in 1921, 46 percent in 1923, and 25 percent in 1924. Notice that the Congress did not cut tax rates until *after* it first shrank expenditures and balanced the budget.

When governments believe they must show that they are “doing something,” a debate arises over the array of policies that should be adopted. Any policy that favors saving over consumption (like cutting taxes on savings, investments, and corporate retained earnings) will shift resources into expediting the recovery. Policies that favor spending at the expense of saving (like stimulus checks) will shift resources away from expediting the recovery—prolonging and possibly deepening the recession.

Furthermore, for businesses, tax cuts are better than stimulus. Suppose that a firm is experiencing a \$2,000 shortfall every month. Assuming constant costs and a gross margin (net sales minus cost of goods sold) of 25 percent, the amount in additional sales needed to cover that shortfall is \$8,000 per month. Turning the business’s malinvestment into a sustainable capital structure will take either \$8,000 of direct *and sustained* monthly government purchases *or* a reduction in taxes of only \$2,000 per month. Thus, in general, the tax-cut approach will place a lower burden on the government (the taxpayer) and still yield the same result.

SUMMATION

The ABCT is a model that shows the cause of the business cycle, the reasons why it affects the *entire* economy, the instability of the artificial boom, the cluster of errors found at the upper turning point of the cycle, and the necessity of the liquidation phase, and that points to time-tested methods for recovery. It is a comprehensive theory that has a beginning, a middle, and an end, and it is also backed by empirical studies.¹⁷ The attempts to refute of the ABCT tend to either criticize the empirical studies or create caricatures of the ABCT. We will explore these criticisms in part 3.

On the other hand, modern macroeconomic theory presents itself in the form of a bicyclist who has been hit by a car. The modern theory is really two separate and independent models. The first model is a theory of a boom and a return, while the second is a theory of a collapse and a return. Modern macroeconomic theory does not really have a

¹⁷Two full-length studies that apply the ABCT to the 2007–9 business cycle are Thomas Woods’s *Meltdown* (2009) and Harry Veryser’s *It Didn’t Have to Be This Way* (2012).

business *cycle* theory. In contrast, the Austrians have a theory that presents a coherent story (which is supported by historical cases), has a defined cause and cure, and can answer several more questions.

The ABCT is not perfect. The model presented above is a simplified version of the theory. When some of these assumptions are relaxed, complications arise. These difficulties are not so much problems with the theory as they are complexities that require further thought and clarification. It is to these unresolved problems that we now turn.

Part 3

Advanced Considerations

IF THE ABCT IS SO CLEAR AND CORRECT, then why haven't more economists adopted this model? There are several reasons why, the first of which is that the ABCT has not been taught in most economics classes. When the ABCT has been presented, it has often been done poorly. For this reason, many economists are simply unfamiliar with the ABCT, while those who have heard of it tend to get much of it wrong. Some think of it as an overinvestment theory, others see it as unable to deal with expectations, and a few find it so much the opposite of what they were taught that it is simply hard for them to accept.

Many economists claim that the ABCT is merely armchair theorizing without empirical data to back it up. While there is some merit to the claim about the data not backing up the ABCT, the problem centers on the data itself and not on the theory.¹ In the 1950s, when many macroeconomic statistics were developed and refined, the dominant macroeconomic model was classical Keynesianism. The statistic on which the most attention was focused was the gross national product (GNP)—now gross domestic product (GDP)—which combines consumption (C), investment (I), government spending (G), and net exports ($X - M$). When GDP increases, these economists say that there is economic growth. When GDP falls, they say the economy is in recession. Unfortunately, these claims oversimplify economic growth. Austrian the-

¹Yeager (1997, 253–56) presents a sympathetic criticism of the ABCT from the perspective of a monetary disequilibrium theorist. He argues that the disturbances caused by monetary injections are insufficient to cause a business cycle.

ory looks deeper than aggregate spending. According to the Austrian model, when people change their time preferences (like in our example where people became more patient), we see that while consumer goods (*C*) and investment goods (*I*) move in *opposite directions*, we remain at full employment on the PPF. The Keynesians (and many modern macroeconomists) claim that the economy is in recession when consumption falls more than investment. In contrast, the Austrians point out that as long as we remain on the PPF, we will remain at full employment and not regress.

As this basic example demonstrates, the statistics are constructed with the classical Keynesian model in mind. In order to empirically verify the ABCT, a new set of statistical aggregates would have to be developed.

UNRESOLVED PROBLEMS WITH THE SOP

The SOP has been presented as a continuous input–continuous output model. Natural resources and labor are combined with capital equipment to produce intermediate capital goods that flow through the SOP. These goods in process are sold to the next stage in the production structure, showing how the partially finished goods move forward and how financial capital moves backward through the SOP. Furthermore, the model shows how this process takes time. Thus, our model of the SOP shows how the original factors of production interact with all the different types of capital to continually produce consumer goods for as long as inputs are fed into it.

The First Challenge: What Is Capital?

Unfortunately, we had to overlook some challenges in simplifying our model. The first challenge is how to define capital and a capital structure. Capital has been defined in many different ways by many different authors. Some say that capital is money. Others say that it is tools, machines, and equipment. Even within the Austrian school, there is no consensus on the most appropriate definition. Carl Menger (1976), founder of the Austrian school, distinguished between the stock of useful capital goods and the flow of their services, arguing for a subjective theory of capital value. Menger's main disciple, Eugen von Böhm-Bawerk (1959), rejected Menger's subjective approach and

focused on the role that time plays in the production of intermediate products. Richard Strigl (2000), on the other hand, conceptualized capital as an aggregate fund for the macroeconomy. Ludwig M. Lachmann (1978), following Hayek, envisioned the economy as a structure of heterogeneous capital goods that could be both complementary to and substitutable with various other capital goods. Israel M. Kirzner (1966) viewed capital as the unfinished intertemporal plans of entrepreneurs. Finally, Peter Lewin and Nicolás Cachanosky (2019, 2021) argue that Mises had a financial view of capital, which they have adopted and extended using a finance tool called duration.² Each author has produced a different way of looking at the economy and thereby has come to a different conclusion. The variation can be extreme. For example, M. A. Abrams (1934) used Hayek's model of the SOP and business cycle to argue in favor of socializing the entire loanable funds market.

The Second Challenge: Valuing Capital

The second challenge is determining the value of capital. Focusing on capital equipment, we can see that there are three generally used methods of determining capital's value.

The first method, the historical approach, uses the price when the piece of equipment was purchased—its book value. Accountants often use the book value of assets in their recordkeeping. Unfortunately, historical values do not contain useful information for entrepreneurs. These historical prices are sunk costs and have no bearing on future market conditions. Projecting historical patterns into the future for the purpose of decision-making assumes that history always repeats itself, which is an unsound conjecture.

The second method uses the current price, the market value of the capital. While this is better than using historical prices, it focuses on today's value and not the future value. What is needed is a forward-looking valuation.

The third method attempts to generate this future-oriented perspective. It is called the discounted-cash-flow approach. The idea is for

²Duration is a tool which determines the number of years (as a weighted average) that are needed to recover the purchase price of a bond based on the pattern of the cash flows over the life of the bond.

an entrepreneur to estimate what the future returns (cash flows) of the asset will be and then discount these returns by an opportunity cost, which is usually compounded. Again, the difficulty lies in the entrepreneur's ability to get these numbers right. Those who estimate them more correctly will net larger profits than those who do not.

The value of a project depends upon the value of the capital equipment, meaning that economists' understanding of how entrepreneurs decide whether to proceed with an investment depends upon their conception of capital value. Economists assume that entrepreneurs use the third method of valuation and that their projections into the future are generally correct. Under this assumption, entrepreneurs decide whether to proceed with a project by comparing its rate of return with the overall rate of return (the slope of the SOP). The equilibrium slope of the SOP thus represents both the rate of return between the various stages of production and the rate of return of the individual projects.

The Third Challenge: Complements and Substitutes

The third challenge regards complements and substitutes. In the mainstream framework, capital is homogeneous and perfectly substitutable. The SOP would be irrelevant if all capital were substitutable. This is, in fact, *exactly* how mainstream economists view capital—they ignore the SOP. However, this perfect-substitutability assumption does not hold up in the real world. Suppose that a baker has two identical delivery trucks. Is the second truck a substitute for or a complement to the first? Since they are identical, they are clearly substitutes. However, the trucks can be on two separate delivery routes at the same time and thereby complement each other as well.

Now let us consider a more difficult case. Are closed-circuit cameras a substitute for a helicopter? Physically, they are nothing alike. The key is in discovering the use of the tools. If someone would like to know what the morning traffic in the city is like, a helicopter can be flown around the city. Alternatively, a series of cameras connected to a central monitor can be placed around the city to easily collect the information. When it comes to discovering what the morning traffic is like, then, a helicopter and closed-circuit cameras are substitutes.

The substitutability or complementarity of inputs is determined by how they are used, not by their physical aspects. In other words, there is no a priori method of determining whether two goods are comple-

ments or substitutes. If we examine the real world, we see that although some capital is substitutable, *most* capital is arranged into *complementary patterns*. The structure of production *represents the degree of capital complementarity in the market*. The challenge for the economist is that the indeterminacy of a good's substitutability or complementarity complicates (or possibly even negates) the graphical presentation of the SOP.

The Fourth Challenge: Dual-Purpose Items

A fourth challenge is dual-purpose items. Some goods have multiple uses. A computer is an excellent example of a good with many uses. A computer is used as a consumer good when games and other fun activities are played. However, the same computer becomes a capital good (input) when it is used to manage a small business's accounting books. Like distinguishing between complements and substitutes, determining whether a good is a consumer good, a capital good at a late stage of production (near the consumer), or a capital good at an early stage of production (near the beginning of the production process) depends upon how the good is used. And like the third challenge, the issue of dual-purpose items makes the graphical representation of the SOP less useful.

On a positive note, an economy whose tools are very flexible will recover from a recession much more quickly than an economy whose tools are not flexible. But the question becomes how to express the degree of flexibility in the SOP. Unfortunately, the answer is that it cannot be modeled graphically. Like all models, the SOP necessarily simplifies economic relations and leaves some issues aside. The graphical model presented above is an analytical tool to help us to organize our thoughts. It gives us a place to begin our analysis and a foundation to build upon.

The Fifth Challenge: Recursive Loops

The final challenge with the SOP resides in recursive loops. In a recursive loop, an input good creates an item that then produces the ingredients for the production of the first item. For example, steel is used to make heavy mining equipment. The mining equipment then extracts iron ore from the ground to make steel, which in turn is used to make more mining equipment. The more recursive loops are embedded in an economy's SOP, the more difficult it is to read its economic data.

UNRESOLVED PROBLEMS IN THE LOANABLE FUNDS MARKET

In part 2, we presented a model of the loanable funds market with a single interest rate. The assumption of a single interest rate significantly simplified the model. There are a plethora of interest rates in the economy. Arrays of interest rates span both numerous risk levels and various maturities. These arrays are respectively called the risk structure of interest rates (RSIR) and the term structure of interest rates (TSIR).

A single interest rate assumes that all projects have the same amount of risk. The RSIR relaxes this assumption and says that some projects are riskier than others. The risk that varies across this array is default risk, the probability that the project will fail and produce no return. The higher the risk of default, of not getting any money out of an investment, the higher the rate of return needs to be to entice investors to accept this risk. Calculating default risk is not an easy task for the entrepreneur. Those entrepreneurs who are able to more correctly forecast risk receive larger returns than those who are less able.

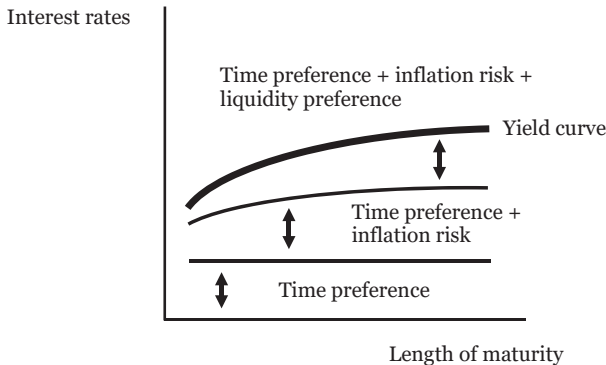
In the model, we also assumed that there is a single interest rate for all maturity lengths. The TSIR relaxes this assumption and allows interest rates to vary across maturities. Some loans are very short, like overnight repurchase agreements. They take place at the end of the business day and are over by the beginning of business the next day. The interest rate for these loans tends to be very low. Other loans last for fifteen or thirty years, like mortgages. The interest rate for these longer-term loans tends to be higher.

There are two reasons why interest rates change as the term of investment changes. The first reason is inflation. If people believe that inflation will persist in the future, then an inflation premium will have to be added to all of the rates. If people expect the inflation rate to accelerate over time, then the inflation premium will also be higher for longer investment maturities.

Secondly, since the future is uncertain, there is a precautionary need for access to liquidity in the future. For example, a person might be comfortable loaning out \$10,000 for six months, but not having access to that money for six years is a different story. The probability of a need for liquidity increases the farther out into the future we look. Thus, not only is a liquidity premium added to interest rates as a loan's

maturity lengthens but the liquidity premium increases as the maturity grows. For example, while a six-month bond might only have a 0.5 percent liquidity premium, a ten-year bond might have a 2.0 percent liquidity premium.

Adding these two factors to time preference produces the TSIR. When the default risk is zero, the TSIR has a special name: the yield curve (figure 18). Since US Treasury securities are considered to have zero default risk, the yield curve is a very popular financial tool and economic indicator. The yield curve *tends* to slope upward as a result of the interplay between its three components—time preference, inflation risk, and liquidity preference—and the arbitrage that takes place between the various maturity segments. This upward slope can change when there are changes in other factors of the economy. The factors that impact the yield curve allow it to function as a forecasting tool for the macroeconomy.



Note: Time preference is presented as being constant across the varying maturities. Some economists, like Rothbard (2009, 449), argue that time preference should be the same across time (flat), while others argue that our impatience should increase as we move further into the future (upward sloping). The reality is that time preference is so intertwined with inflation risk and liquidity preference that it can only be conceptually separated from them. No praxeological or empirical test could determine which interpretation of time preference is correct. As a result, there is no a priori answer, and the shape of the time preference curve is a result of one's assumptions about human behavior.

Figure 18: The components of the yield curve

One significant error made by economists has been to associate a particular segment of the SOP with a particular segment of the yield curve. Suppose that an entrepreneur has a plan that will take three years to show any return. To finance this venture (financing always comes from savings—deferred consumption), the entrepreneur’s own savings, someone else’s savings, or both must be used. The options available to the entrepreneur-borrower are plentiful. The borrower could match maturities and take out one three-year loan, or the borrower could obtain three one-year loans, or two eighteen-month loans, or one two-year loan to be followed by a one-year loan, and so forth. Here is the most important part: the *maturity of the loan* (i.e., where it is in the TSIR) and the *location* of the business within the structure of production have *no direct time link* to each other. Figure 19 illustrates the error of assuming this direct time link. Any conclusions that stem from this faulty analysis are illegitimate.

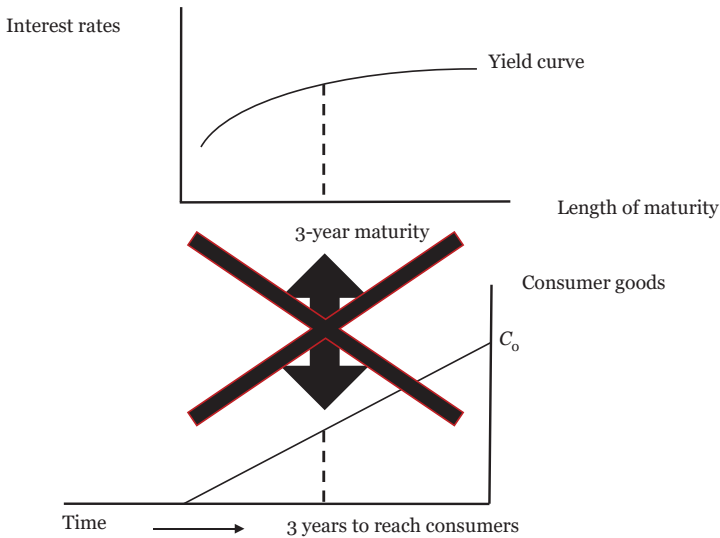


Figure 19: No direct time link between the yield curve and the SOP

UNRESOLVED PROBLEMS WITH THE ABCT

In our simple ABCT model, we assumed that monetary expansion lowers the interest rate. When we relax the assumption of a single interest rate, we can maintain the conclusion that credit expansion affects all maturities in the TSIR and in the RSIR. However, we do not know if this decrease in rates caused by monetary expansion will be evenly applied across each array of interest rates. Nevertheless, we can state that this change stimulates both consumption and investment.

Entrepreneurs use tools like net present value (NPV) in deciding whether to accept or reject an investment project. The NPV is the summation of appropriately discounted projected future returns (cash flows). The discount rate often used is the weighted average cost of capital (WACC). The WACC is a weighted average of the opportunity costs of the different types of financial capital: equity, debt, and preferred stock. A company's opportunity cost of equity (RE) is the rate of return investors expect to receive from their equity investment in a company, a company's opportunity cost of debt (RD) is the rate of return investors expect to receive from their purchase of a company's bonds, and a company's opportunity cost of preferred stock (RPS) is the rate of return investors expect to receive from their investment in a company's preferred stock.³ Thus the $WACC = [WE \cdot RE] + [WD \cdot RD (1 - \text{corporate tax rate})] + [WPS \cdot RPS]$, where WE , WD , and WPS are the respective weights.

When there is monetary expansion, there is a change in the WACC, which promulgates a rebalancing between equity and loans. In general, the WACC falls as the opportunity cost of debt (RD) falls, and this has a greater impact on the (projected) longer-lived fixed capital goods (Cwik 2008). When longer-lived fixed capital goods are encouraged, more roundabout (complex) capital structures are undertaken throughout the SOP. More specifically, projects with longer maturities and more specific capital equipment are encouraged. The key point is that during the unsustainable boom, there is a switch from more substitutable capital to more complementary (specific) capital, making the liquidation phase deeper and more prolonged.

³Not all companies utilize all three types of financial capital.

In our model, we saw that as the artificial boom closed and the economy neared the upper turning point, all the entrepreneurs scrambled to find bricks (real resources) to finish their projects. When the assumption of a single interest rate is relaxed, we see that there is in fact more pressure on short-term interest rates than on long-term interest rates during the upper turning point. The reason is simple: if entrepreneurs are not able to get any financing for the needed bricks, they cannot complete and sell the houses they have invested in building. If they do not sell the houses, they lose all that they have sunk into the project. Thus, they are willing to borrow at high short-term interest rates to prevent big losses. The result of this scramble for short-term funds is seen in the empirical relationship of an inverted yield curve (Cwik 2005). Historically, an inverted yield curve has preceded a recession by an average of four to six quarters.

Conclusion

IN PART 1, WE EXAMINED ECONOMIC growth and how an economy “goes right.” The “magic” formula for economic growth is not magic. The economy does not grow automatically. It requires the coordination of millions and millions of people every day.

A key aspect to the proper working of an economy is the price system. Prices perform an incredibly valuable economic function. Prices are packets of information. They communicate, to anyone who wishes to look, the relative scarcities of the various goods and services available. When something changes in the economy, prices transmit this information to anyone and everyone. When a new deposit of a resource is located, the lower price informs everyone of this discovery. When a storm disrupts supply lines, higher prices tell users to be more cautious with goods and suppliers to rush more to the affected area. When a business finds a new way to be more efficient, its new prices communicate this new reality to all.

Everyone, from entrepreneurs to consumers to employees, uses prices to integrate their personal plans with the greater whole. Should we invest in project A or project B? Should I buy that product now or look for something else? Should I go to work today or quit and find a new job? Each of these questions is unanswerable without the price system. The entrepreneur can calculate the different rates of return for the projects and decide. The consumer can see that the price is higher than acceptable and look for something else. The employee can see that there is another job that offers a higher wage. Prices are not a perfect reflection of our subjective preferences and are constantly changing, however, they serve the economic

function of communicating important knowledge quickly and efficiently. There is no substitute.

In part 2, we saw the effects of manipulating prices. When the government imposed a price cap on the loanable funds market, a political purpose was served. When the central bank chose the path of monetary expansion, it decided that it could accelerate the growth of the economy. As economists, we need to ask, “Are these policies enhancing or detracting from the market’s ability to efficiently and effectively serve consumers? Do these policy changes enhance an economic function? Do the new prices that result from price caps or expansionary monetary policy communicate new scarcity ratios between goods and services? Do they reflect new tastes and preferences? Have expectations about the future changed?” If the new prices do not reflect the underlying economic reality, they are disruptive and lead away from economic stability.

Governments often intervene in markets. When small-scale interventions occur, the effects tend to be limited in both time and place. However, when the intervention is systemic and persistent, it leads to the ultimate disruption of the economy—a business cycle. The best way to avoid such a calamity is by not starting down the path of monetary expansion. Knowing how business cycles get started helps to give us the strength to resist the siren song of easy money. In this vein, let us close with Rothbard.

The time is ripe—for a rediscovery, a renaissance, of the [ABCT]. It can come none too soon: if it ever does, the whole concept of a Council of Economic Advisors would be swept away, and we would see a massive retreat of government from the economic sphere. But for all of this to happen, the world of economics, and the public at large, must be made aware of the existence of an explanation of the business cycle that has lain neglected on the shelf for all too many tragic years.¹

¹Rothbard (1996, 91).

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